

RDS (RELATIONAL DATABASE SERVICE)

RDS (Relational Database Service) is a fully managed database service provided by Amazon Web Services that allows users to create, operate, and scale relational databases in the cloud without managing the underlying infrastructure.

With Amazon RDS, AWS takes care of:

- Server setup and configuration
- Database installation
- Automatic backups
- Software patching and updates
- Scaling and storage management
- High availability (Multi-AZ deployment)

Supported Database Engines

Amazon RDS supports multiple relational databases such as:

- MySQL
- PostgreSQL
- MariaDB
- Oracle
- Microsoft SQL Server
- Amazon Aurora

Key Features

- Automated Backups – Data is backed up automatically
- Scalability – Easily increase storage and performance
- High Availability – Multi-AZ deployments for fault tolerance
- Security – Integrated with IAM and VPC
- Monitoring – Works with CloudWatch

Master (Primary)

- The Master database is the main database
- Handles all write operations (INSERT, UPDATE, DELETE)
- Also handles read requests

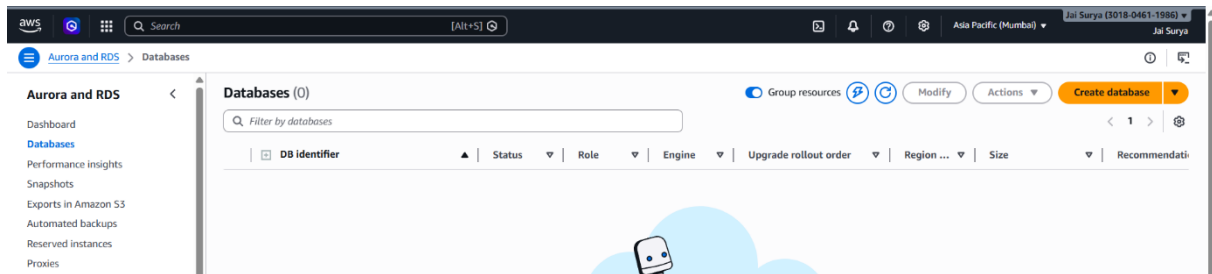
Slave / Replica?

- A Slave (Replica) is a copy of the master database
- It receives data from the master using replication
- Mainly used for read operations

HANDS-ON FOR CREATING RDS AND CONNECTING MYSQL SERVER

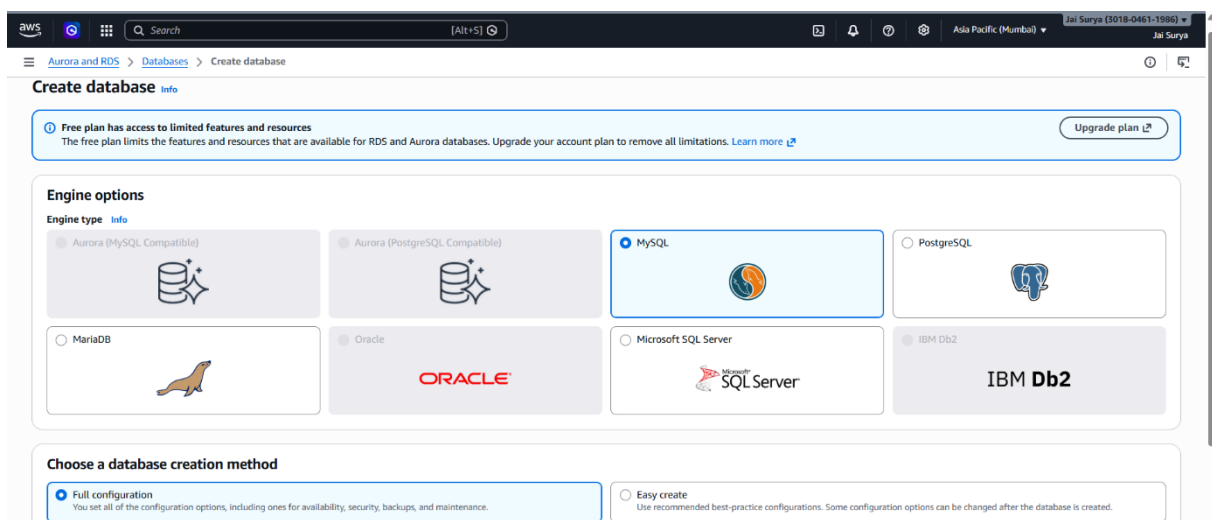
Screenshot: Create RDS Database

- Go to RDS Dashboard
- Select Aurora and RDS
- Click on Create database



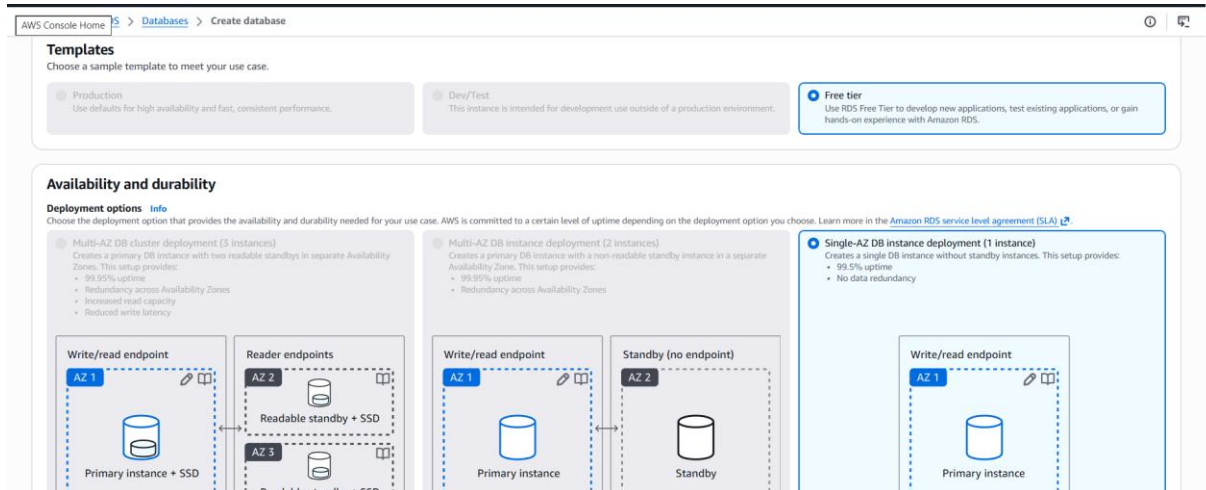
Screenshot: Select Database Engine

- Select MySQL as the database engine
- Choose Full configuration option



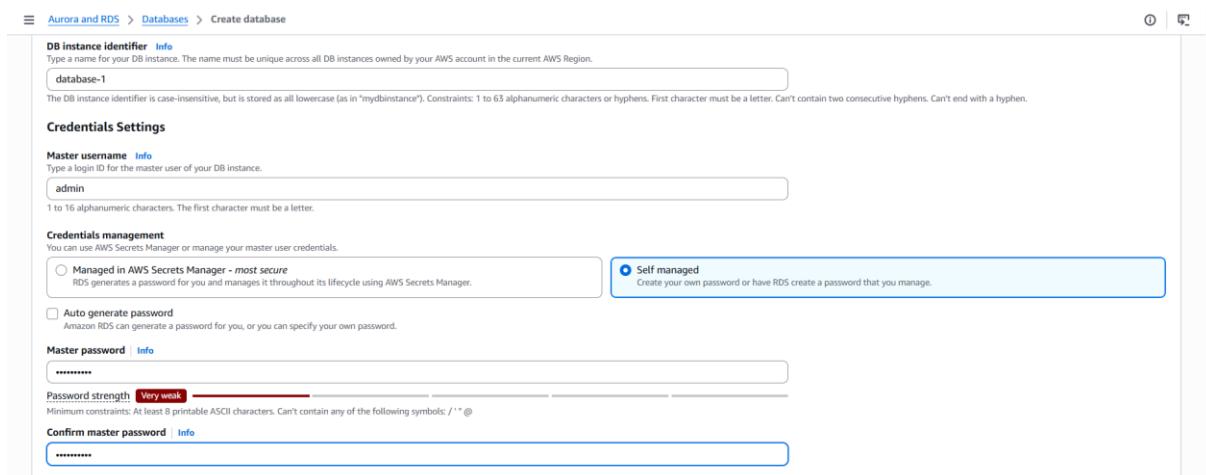
Screenshot: Choose Deployment Type

- Select Free tier option
- Choose Single-AZ DB deployment



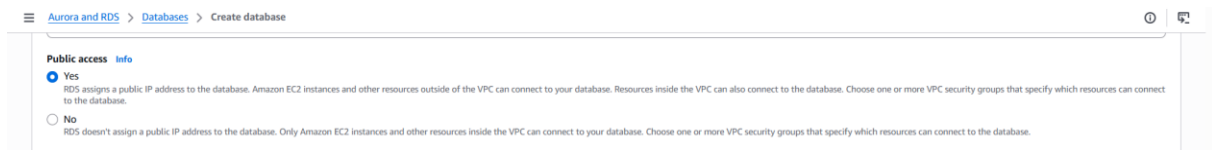
Screenshot: Configure Database Credentials

- Enter database name
- Enter master username
- Select self managed credentials
- Enter master password
- Confirm password



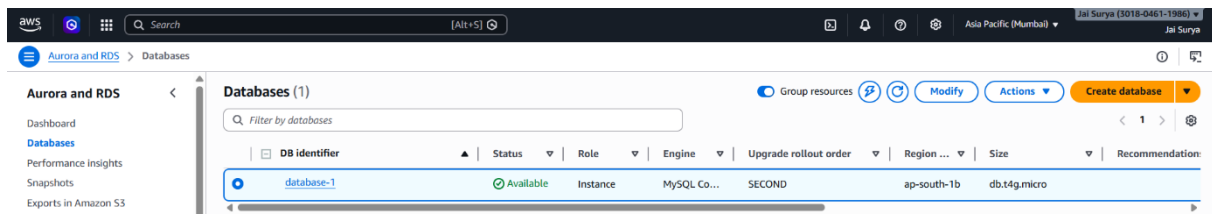
Screenshot: Configure Connectivity

- Set Public access to Yes



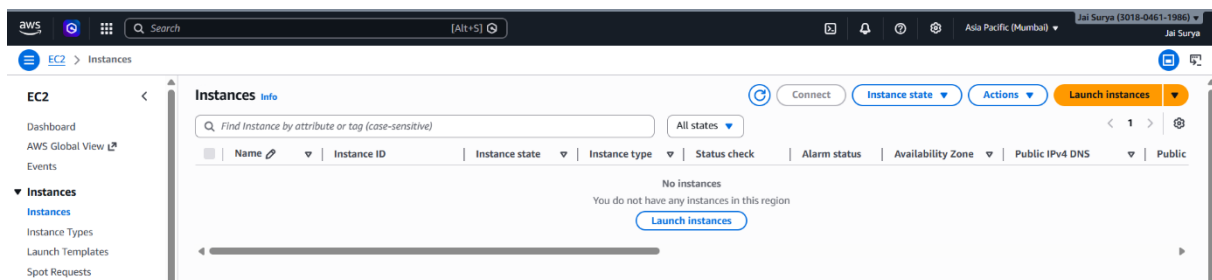
Screenshot: RDS Database Created

- Verify that the master database is created successfully
- Ensure the database status is available



Screenshot: Create EC2 Instance

- Go to EC2 Dashboard
- Click on Instances
- Click on Create instance



Screenshot: Configure Instance Details

- Enter instance name
- Select Ubuntu as the OS
- Select a key pair

EC2 > Instances > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

▼ Summary

Number of instances Info

1

Software Image (AMI)
 Amazon Linux 2023 AMI 2023.10....[read more](#)
ami-0f559c3642608c138

Virtual server type (instance type)
 t3.micro

Firewall (security group)
 New security group

Storage (volumes)
 1 volume(s) - 8 GiB

Cancel [Launch inst](#) [Preview](#)

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

3pmbatch2 [Create new key pair](#)

Screenshot: Ubuntu Instance Created

- Verify that the Ubuntu instance is created successfully
- Ensure the instance is in running state

aws Search [Alt+S] Asia Pacific (Mumbai) Jai Surya (3018-84511586)

EC2 > Instances

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states Connect Instance state Actions Launch instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	rds-ubuntu	i-08a4ed1d73191aa4c	Running	t3.micro	3/3 checks passsec	View alarms +	ap-south-1a	ec2-13-201-120-29.ap-...	13.201

Screenshot: Connect to Instance

- Select the instance
- Click on Connect
- Use SSH or EC2 Instance Connect to connect to the instance

```
aws
Search [Alt+S]
Memory usage: 24%    Users logged in: 0
Swap usage: 0%      IPv4 address for ens5: 172.31.44.240

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-44-240:~$
ubuntu@ip-172-31-44-240:~$
ubuntu@ip-172-31-44-240:~$
ubuntu@ip-172-31-44-240:~$
ubuntu@ip-172-31-44-240:~$
ubuntu@ip-172-31-44-240:~$
ubuntu@ip-172-31-44-240:~$

i-08a4ed1d73191aa4c (rds-ubuntu)
PublicIPs: 13.201.120.29  PrivateIPs: 172.31.44.240
```

Screenshot: Switch to Root User and Install MySQL

- Execute sudo su
- Execute apt update -y
- Execute apt install mysql-server -y

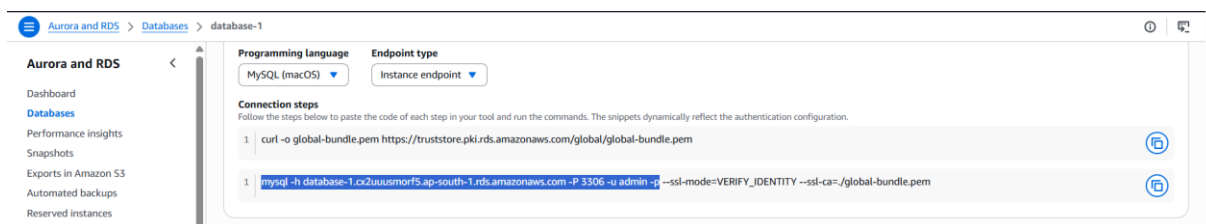
```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-44-240:/home/ubuntu# history
 1  apt update -y
 2  apt mysql-server
 3  apt install mysql-server
 4  history
root@ip-172-31-44-240:/home/ubuntu#
```

i-08a4ed1d73191aa4c (rds-ubuntu)

PublicIPs: 13.201.120.29 PrivateIPs: 172.31.44.240

Screenshot: Copy Database Endpoint

- Go to the created RDS database
- Copy the connection endpoint link



Screenshot: Connect to RDS Database

- Paste the database endpoint in the Ubuntu terminal
- Enter the password to connect to the database

```
root@ip-172-31-44-240:/home/ubuntu# mysql -h database-1.cx2uuusmorf5.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
```

Screenshot: MySQL Connected Successfully

- Verify that MySQL is connected successfully to the RDS database

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 32
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

i-08a4ed1d73191aa4c (rds-ubuntu)
PublicIPs: 13.201.120.29 PrivateIPs: 172.31.44.240

Screenshot: Execute MySQL Queries

- Execute MySQL queries to create and manage database tables
- Verify that queries are executed successfully

```
mysql> create database jai;
Query OK, 1 row affected (0.05 sec)

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| jai |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.02 sec)

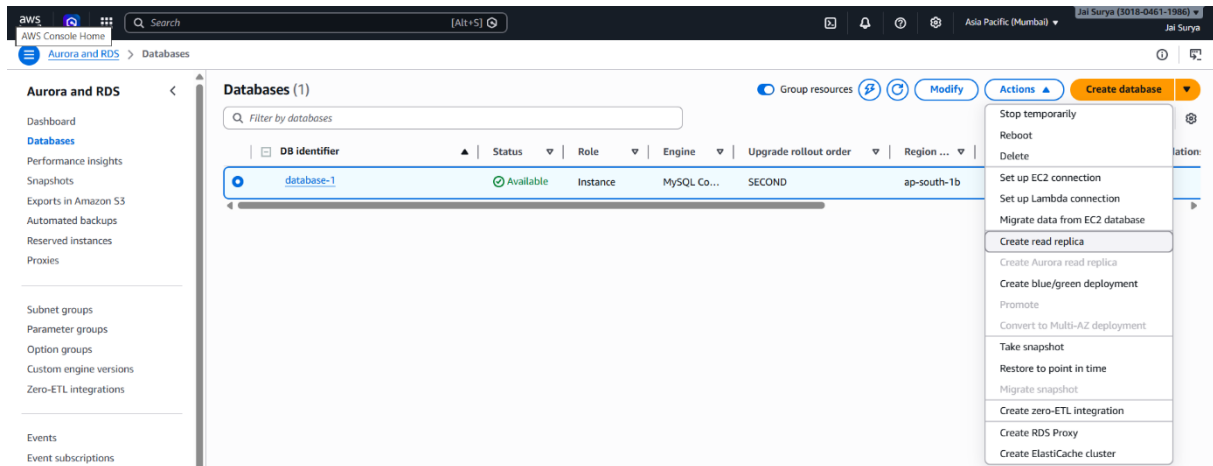
mysql> use jai;
Database changed
mysql> create table details;
ERROR 4028 (HY000): A table must have at least one visible column.
mysql> create table details(name varchar(30),ph int,address varchar(30));
Query OK, 0 rows affected (0.08 sec)

mysql> desc details;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name | varchar(30) | YES | | NULL | |
| ph | int | YES | | NULL | |
| address | varchar(30) | YES | | NULL | |
+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

i-08a4ed1d73191aa4c (rds-ubuntu)
PublicIPs: 13.201.120.29 PrivateIPs: 172.31.44.240

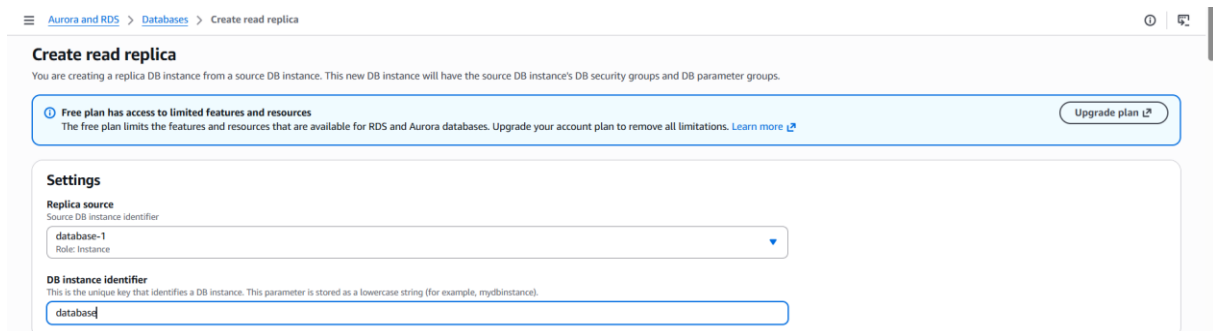
Screenshot: Create Read Replica

- Go to RDS Dashboard
- Select the master database
- Click on Actions
- Select Create read replica



Screenshot: Configure Read Replica

- Enter the read replica name



Screenshot: Read Replica Created Successfully

- Verify that the read replica is created successfully
- Ensure the replica status is available

